

Environmental System Map: Urban Municipal Solid Waste Management

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Deliverable: Module 1 – Environmental System Map

1. Selected Environmental System

The chosen environmental system is the **Urban Municipal Solid Waste (MSW) Collection and Processing System** in a growing tier-2 urban municipality. This system encompasses household waste generation, community segregation, municipal transport, sorting facilities, and final disposal or recycling loops.

2. System Inputs

- **Raw Materials & Resources:** Organic household waste, recyclable plastics, paper, glass, metals, and hazardous household materials.
- **Energy & Fuel:** Fossil fuels (diesel/petrol) consumed by municipal collection trucks and electricity used at material recovery facilities (MRFs).
- **Human Capital:** Sanitation workers, facility operators, collection drivers, and municipal administration personnel.
- **Information & Infrastructure:** Waste bin infrastructure, collection schedules, route-planning software, and municipal environmental policies.

3. Core System Processes

Process Stage	Description
Generation & Segregation	Households and commercial entities discard waste, ideally separating it into wet (organic) and dry (recyclable) streams at the source.
Collection & Transport	Municipal fleet trucks collect waste from residential zones and transport it to transfer stations or centralized processing plants.
Sorting & Recovery	Materials are sorted manually and via automated mechanical separators (magnetic belts, air classifiers) at Material Recovery Facilities (MRFs).
Treatment & Disposal	Organic waste undergoes composting or anaerobic digestion; recyclables are baled for recycling industries; non-recyclable residual waste is sent to engineered landfills.

4. System Outputs

- **Valuable Outputs:** Recycled raw materials (plastics, metals, paper), organic compost for agricultural use, and biogas/energy generated from waste digestion.
- **Unintended Byproducts & Emissions:** Greenhouse gases (methane and carbon dioxide from decomposing organic waste), leachate (contaminated liquid runoff), and vehicular carbon emissions from transport fleets.

5. Identified Inefficiencies

- **Low Source Segregation Rates:** Mixed waste generation contaminates recyclables, increasing the labor and energy required for sorting at recovery facilities.
- **Suboptimal Routing:** Fixed collection schedules without real-time fill-level sensors lead to fuel wastage and unnecessary truck deployment.
- **Organic Waste Loss:** A large fraction of biodegradable waste ends up in landfills rather than composting or biogas plants, generating harmful methane emissions.

6. Future Improvement Opportunities

- **Smart IoT Bin Integration:** Deploying fill-level sensors on community bins to optimize collection vehicle routes and reduce fuel consumption.
- **Decentralized Composting Hubs:** Establishing neighborhood-level composting units to divert organic waste at the source and reduce municipal transport burdens.
- **Enhanced Public Awareness Campaigns:** Implementing reward-based digital programs to incentivize household source segregation.